PLAN · ZENON NETWORK OF MOMENTUM · 11 SEP 2026

Mainnet-Adjacent State Machine

Four proposed ways to run general-purpose contracts next to Zenon mainnet, weighed against what comparable chains actually shipped and against what mainnet allows today. It adds the routes the list was missing, and gives a phased plan for the one it recommends.

MEASURED VALUES: MAINNET RPC + THIS TREE, 11 SEP 2026
INPUTS: PROJECTS/ZENOGLYPHS/RESEARCH · GO-ZENON (TESTNET @ 590F43C)

RECOMMENDED OPTION 4 · ZENOGLYPHS-LITE

Build **zenoglyphs-lite as a based, sovereign EVM rollup**: NoM orders and stores every transaction, anyone can derive the state, and a stock EVM client executes it. Keep zenoglyphs beside it as the asset layer. Keep the EVM-on-zenoglyphs prototype on devnet as the testbed it has been, and don't ship it to mainnet. Don't build the optimistic rollup: NoM has nothing that can adjudicate a fraud proof, and no spork can currently add one.

- **It needs no protocol change, and right now none can be made.** The draft governance ZIP says the location of the genesis spork key is unknown. The community spork key was valid only until momentum 13,243,712 (16 Apr 2026), and mainnet is at 14,164,216. Anything that needs a spork has no date.
- **Nothing is on mainnet yet.** Neither zenoglyphs nor its EVM has a mainnet start height, so choosing a different envelope today costs a spec, not a migration. Facet made the same switch after launch and had to re-execute its old history into the new chain's genesis.
- **It's where the closest precedents ended up.** Ethscriptions became Facet, and KRC-20's team built Kasplex L2. Both are EVMs whose order and data come from the L1, and both left the inscription envelope behind to get there. BRC2.0 kept its envelope because the tokens it had to make programmable already lived inside it.
- **It's cheaper, and it scales where the prototype can't.** An ERC-20 transfer costs 25% less plasma without the envelope. A batch lane carries about 89 signed transfers per L1 block, which a one-inscription-per-call design can't do at all.

THE ONE DECISION THIS TURNS ON

Do zenoglyph assets need to move into contracts *and back out*? If they do, the EVM and zenoglyphs have to become one state machine. That's BRC2.0's answer, and option 3 is the right base, rebuilt on a real EVM (Path B). If one-way deposits, or tokens issued natively as ERC-20/721, are enough, keep the two apart. That's Facet's and Kasplex's answer: option 4 (Path A). This plan assumes Path A, because no zenoglyph asset exists on mainnet yet.

MEASURED · NODE.ZENONHUB.IO:35997 · GO-ZENON SOURCE

What mainnet allows today

Every option has to live inside these facts. Mainnet runs the original 2021 rules, and it is almost empty.

FRONTIER MOMENTUM

14,164,216 · v1

Version 1 means dynamic plasma is not active. There's no price and no fee market, and blocks enter first-come.

`ledger.getFrontierMomentum`

LOAD, LAST 8,640 MOMENTUMS

0.09 blocks / momentum

817 account blocks in 27 hours. 8,315 momentums (96%) were empty. The mean interval was 11.25 s.

`ledger.getMomentumsByHeight`, sampled at the tip

CEILING PER MOMENTUM

100 account blocks

A count, not a byte budget. A design that spends one block per transaction tops out at about 9 tx/s even if it had all of NoM.

`chain/account_pool.go` · `MaxAccountBlocksInMomentum`

DATA PER BLOCK

16,384 B · 21,000 + 68/B

Plasma is 21,000 per block plus 68 per data byte. The flat term is worth about 309 bytes, so the block count matters more than the bytes.

`vm/constants/plasma.go`

PLASMA SUPPLY

1 QSR = 2,100 plasma

Fused QSR, capped at 5,000 QSR (10.5 M plasma) per account. It comes back when blocks confirm, so the tank refills every momentum. That makes it a rate limit, not a fee.

`vm/plasma.go` · `AvailablePlasma`

PROOF OF WORK INSTEAD

≤ 94,500 plasma

That's at most 1,080 data bytes per block, and 141.75 M hashes at the cap: about two minutes at the ~1 MH/s solzen measured in Go.

`MaxPoWPlasmaForAccountBlock`

ORDER INSIDE A MOMENTUM

address ‖ height ‖ hash

Blocks are sorted by the sender's address bytes, not by arrival. Any based design inherits this, so a vanity key with leading zeros gets priority.

`chain/nom/momentum_content.go`

SPORKS

frozen

6 sporks were created and 3 activated (AcceleratorZ, HTLC, Bridge). "Governance Contract" and "Point Timelock Contract" were created and never activated. The community key's window was momentums 10,109,240–13,243,711, and it has closed.

`embedded.spork.getAll` · `vm/embedded/definition/spork.go`

BRIDGE

live · 2 networks

Ethereum (1) and BNB Chain (56). The bridge isn't halted, and the orchestrators treat 90 momentums as final. The administrator adds an EVM network with `SetNetwork`, which needs no spork.

`embedded.bridge.getAllNetworks` · `getOrchestratorInfo`

HISTORY

never pruned

go-zenon keeps every account block's data. Rollup inputs live on every full node forever. That's what makes a based rollup's data available, and it's also a disk cost those nodes pay.

no pruning path in go-zenon; devnet: ~2.1 KB per block per node

⚠ THE DEVNET NUMBERS DON'T CARRY OVER

The benchmark in `research/evm-benchmark.md` ran under dynamic plasma: 4.2 M base plasma per momentum, 42.4 M per account, and the wedging and rebids it found. Mainnet runs the older rules: a flat 100 blocks per momentum, 10.5 M plasma per account, and no price at all. Mainnet is also about 99.9% idle, so capacity won't decide this. Trust, exits, tooling and demand will.

WHAT OTHERS SHIPPED · CHECKED 11 SEP 2026

What comparable systems chose

Each of the four proposed options has been tried by someone, including twice inside Zenon. This is what they chose, and what happened to them.

SYSTEM	CLOSEST TO	WHAT IT CHOSE	VALUE IN AND OUT	HOW IT WENT
BRC2.0 Bitcoin · live Sept 2025, block 912,690	Option 3	An EVM (revm) inside the BRC-20 indexer. Contracts are deployed and called by inscription (`"p":"brc20-prog"` , ops `deploy` / `call` / `transact`). The gas budget is 12,000 per inscribed byte, and precompiles read Bitcoin transactions and BIP-322 signatures.	BRC-20 in: transfer to `OP_RETURN "BRC20PROG"` . Out: a withdraw inscription. No BTC peg.	Shipped, and made existing tickers (ORDI, SATS) programmable in phase 2. It later added `transact` for pre-signed EVM txs and a compressed base64 field. Those are its answers to option 3's two weaknesses: wallets and header bytes.
Facet Ethereum · VM Nov 2023 → EVM Dec 2024	Option 3, then option 4	It began as "Dumb Contracts", a Ruby VM on Ethscriptions, then rebuilt as a based rollup. Transactions are type `0x46` , `rlp([chain_id, to, value, gas_limit, data, mine_boost])` , sent to `0x…face7` with no L2 signature: the L1 sender is `msg.sender` . facet-node drives an op-geth fork over the Engine API.	Gas token minted from L1 calldata gas burned. No canonical bridge; independent bridges use ZK fault proofs (OP-Succinct).	L2BEAT Stage 2. It passes the walkaway test, and total value secured is about $650K. Its genesis re-executed the legacy Dumb Contracts history. A sound design didn't bring adoption.
Kasplex L2 Kaspa · Aug 2025	Option 4	A based EVM rollup from the team behind KRC-20 inscriptions (130 M+ txs). EVM bytecode rides Kaspa transaction payloads, and the DAG's order is the rollup's order.	Bridged KAS pays gas. A relayer turns Ethereum-format txs into Kaspa txs, so MetaMask works. KRC-20 crosses over by a multisig bridge.	Live. The inscription team kept KRC-20 and the EVM as separate machines, joined by a bridge. That's Path A.
Igra Kaspa · Mar 2026	Option 4 plus attesters	A based rollup with a Reth-derived EVM. Attesters check execution against Kaspa's order, and Sigma Prime audited it.	KAS locked at an L1 entry address mints iKAS 1:1. iKAS burns to redeem.	Live. Kaspa's Toccata hard fork (30 Jun 2026) then added covenants, a ZK-verifier opcode and sequencing commitments. That is the L1 upgrade based rollups need for trustless exits.
Counterparty EVM Bitcoin · 2014–2016	Option 3	It ported pyethereum into the Counterparty indexer. A 2016 mainnet proposal came with a kill switch in case of a critical bug.	XCP inside; no BTC peg.	Put on hold by 2017. The forum's verdict: "Projects that need this functionality will naturally gravitate to Ethereum."
Supernova Zenon · Aug 2024	Option 1 as a sidechain	A Cronos-based (Cosmos SDK) EVM chain run by pillars as its validators, the "Extension Consensus Group". Its own consensus, not NoM's.	xZNN over the orchestrator bridge, with a 1% exit fee to those pillars. The planned trustless version needed an NoM light client and an embedded contract, which means a spork.	Dormant. The repo was last pushed on 24 Dec 2024. Its RPC (`supernova.zenon.red`) and explorer (`novascan.io`) no longer resolve, and its history can't be re-derived from NoM.
Project Zeno Zenon · spec 2025–26	Option 1, bonded rather than optimistic	A "Domain Settlement Layer": NoM orders the inputs, one bonded executor runs WASM off-chain, and a new Settlement embedded contract records commitments.	Custody is capped per domain, with withdrawal delays and value caps. Fraud proofs come in phase 2, validity proofs later.	Spec only. The Settlement contract needs a spork, so it's blocked today, and the `go-executor` repo is still empty. Its own docs describe phase 1 as "bonded attestation with bounded custody, not fully trustless execution".

Also relevant: Alkanes (Bitcoin, Jan 2025) runs WASM contracts on top of protorunes. That's option 3 with WASM in place of the EVM, and indexer maturity was its recurring concern. IOTA ran committee-validated EVM chains anchored to its DAG from May 2024, then rebuilt its L1 on MoveVM in 2025.

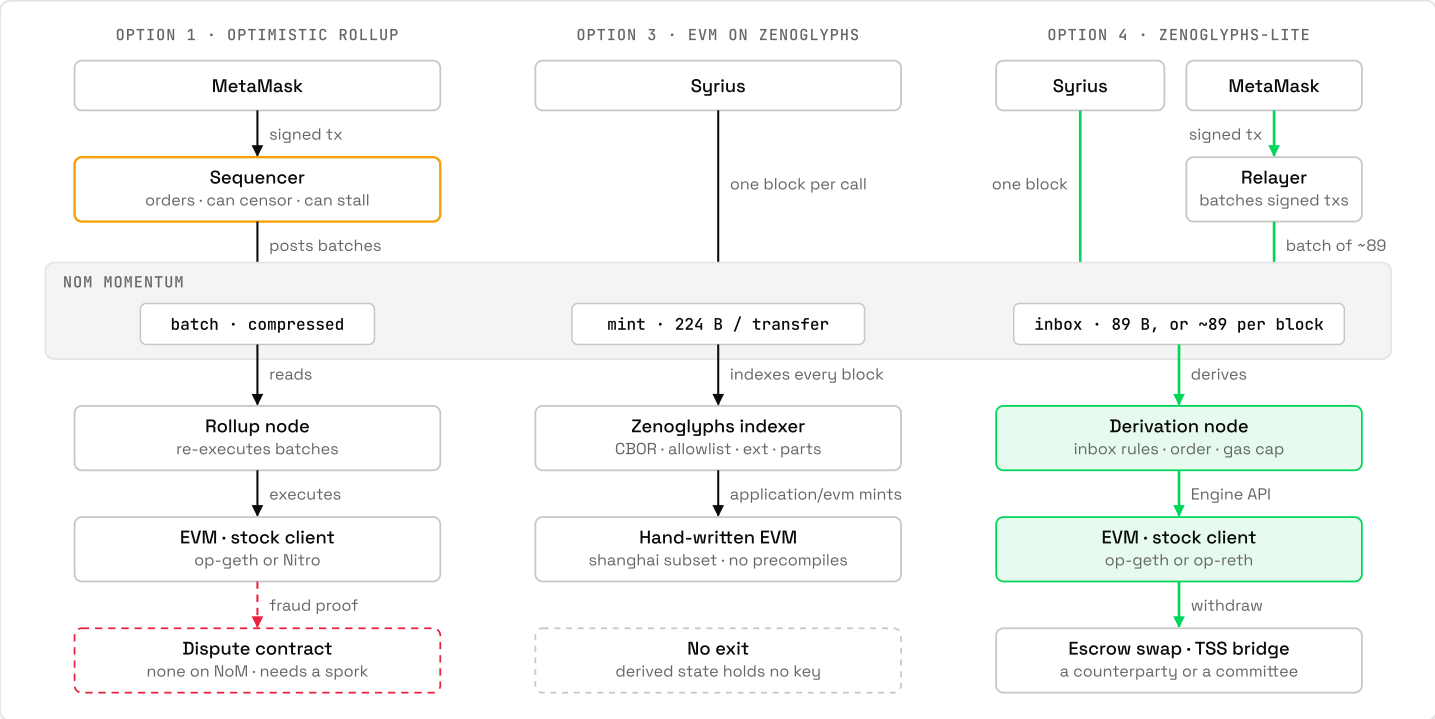
- **An inscription protocol that adds contracts either merges with its assets or separates from them.** BRC-20 merged, so the EVM runs inside the indexer. Ethscriptions and KRC-20 separated, into a lean-envelope chain. The deciding question was whether existing assets had to become programmable in place.
- **Nobody got trustless exits without help from the L1.** Facet has Ethereum contracts. Kaspa's rollups waited for Toccata. Supernova, IOTA and Project Zeno phase 1 use committees or bonds. NoM can't give that help today.
- **Committee chains die with their committees. Derived chains survive their operators.** Supernova's history left with its RPC. A chain derived from NoM can always be rebuilt by anyone from NoM itself.

- **Adoption followed assets and wallets, not architecture.** BRC2.0 launched on tokens people already held, and Kasplex and Igra shipped with MetaMask. Facet's Stage 2 design holds $650K. Counterparty's EVM never found a reason to exist.

THE FOUR PROPOSED OPTIONS

The options, weighed

Options 1, 3 and 4 all put an EVM next to NoM. The figure shows where they actually differ: who orders a call, how many rule sets sit between NoM and the state, and whether anything can take value back out.



How a call becomes state. Option 1 puts a trusted operator (amber) before NoM, and its safety depends on a dispute contract NoM cannot host (red, dashed). Option 3 sends every call through two rule sets, zenoglyphs' and then the EVM's, one call per L1 block. Option 4, in green, has one rule set, a stock EVM, and a batch lane that lets MetaMask users in through a relayer.

OPTION 1

✕ NOT AS SPECIFIED

Optimistic rollup sidechain

On NoM, this is a sequencer that orders EVM transactions and posts compressed batches to NoM as data, plus a proposer that posts state roots anyone may challenge within a window.

FOR

- Off-the-shelf stacks (OP Stack, Arbitrum Nitro) bring full EVM equivalence: MetaMask, Foundry and Blockscout on day one.
- The sequencer gives soft confirmations in under a second, instead of waiting ~11 s for a momentum.
- It's the densest use of NoM. Compressed batches spread the 21,000-plasma block cost over hundreds of transactions.

AGAINST

- "Optimistic" assumes an L1 contract that holds deposits and settles challenges. NoM has no general VM to run the dispute game, and adding one is a spork that can't currently be activated.
- Without that contract it becomes something else: a sequenced sovereign rollup, where challenges persuade nodes rather than a contract, or a sidechain whose peg is a committee. Zenon has already run the second. Supernova is dormant.
- The sequencer is one operator that can censor, reorder and stall. Forced inclusion through NoM has to be designed in from the start.
- It's the most engineering. op-node's derivation, the batcher and the L1 contracts all assume Ethereum.

CAN'T DO Enforce a fraud proof on NoM, or release L1 ZNN without a committee.

VERDICT Don't build it as specified. Its useful parts (compressed batches, a sequencer for fast pre-confirmations, forced inclusion) can be added to option 4 later, without the L1 contract it can't have.

OPTION 2

✓ KEEP · ASSET LAYER

Zenoglyphs, the inscriptions metaprotocol

Already built. The v1 wire format is frozen with golden vectors, and there are seven opcodes: mint, transfer, multi-part, collection, and three for dispensers. The indexer has checkpointed reorg recovery and cross-indexer state roots, and there's a viewer and a Telegram front end. It isn't on mainnet yet: the start height is unset.

FOR

- No protocol change. Pillars see ordinary 0-amount sends with data, which they already relay.
- Dispensers already settle against real ZNN or ZTS paid on L1. That's the pattern trust-minimized exits reuse later (route D).
- Content, collections and composites are what people will actually see and hold.

AGAINST

- It's a fixed-function state machine. Every new behaviour is a new opcode or a version bump.
- Nothing general-purpose can be built on it without a second machine next to it.

CAN'T DO Run arbitrary logic. That's by design, not a gap.

VERDICT Launch it as the asset layer. The general-purpose machine should sit beside it, not inside it (unless you choose Path B).

OPTION 3 · BUILT ON DEVNET, 9–10 SEP 2026

⚠ DEVNET TESTBED

An EVM that lives on top of zenoglyphs

Every deploy and call is a zenoglyphs mint with mime `application/evm`, and the operation sits in `ext`. State comes from replaying those mints in canonical order. The fee has been live on devnet since momentum 55,100, and a stock solc 0.8.26 ERC-20 runs unmodified.

FOR

- It already works end to end on devnet, including publishing from Syrius and an explorer.
- A contract's address is a commitment to its code (`0x16 ‖ keccak(assetId)[13:]`), so the address names an inscribed artifact anyone can fetch.
- The only zenoglyphs consensus change was one allowlist entry.
- It has a direct precedent that shipped: BRC2.0 is this shape.

AGAINST (MEASURED ON 8,464 DEVNET CALLS)

- 156 bytes of envelope wrap a median 36 bytes of calldata. An ERC-20 transfer carries 224 bytes and costs 36,232 plasma, where 89 bytes would do.
- Every call mints an asset: about 1.56 KB of indexer rows per call, or 44 bytes stored per byte of calldata.
- Its consensus is coupled to zenoglyphs'. EVM validity inherits canonical CBOR, the allowlist, multi-part chaining and the `ext` bounds. Zenoglyphs defines `ext` as consensus-inert, and the EVM makes it consensus-critical, so a zenoglyphs change can fork the EVM without touching it.
- One call per L1 block, authenticated by the block's own signature. That rules out batching, caps it at ~9 calls/s even with all of NoM, and locks out MetaMask.
- A hand-written interpreter carries the consensus risk. It has no CREATE/CREATE2, DELEGATECALL or precompiles (so no `ecrecover`, and the bridge's own Solidity can't run), no transient storage, a flattened gas schedule, and a non-standard state root.

CAN'T DO Batch. Take MetaMask-signed txs without a BRC2.0-style `transact` op. Move zenoglyph assets back out of contracts unless the zenoglyphs indexer itself executes the EVM.

VERDICT Keep it on devnet as the testbed. Take it to mainnet only on Path B, and then on revm or geth's `core/vm` in place of the interpreter, with a `transact` op and the S1 items in `evm-todo.md` closed.

OPTION 4 · NOT BUILT YET

✓ BUILD THIS

Zenoglyphs-lite: an EVM inscribed without the envelope

Blocks sent to one inbox address carry an EVM transaction directly: no CBOR, no mint. The inbox sits in the `0x16` namespace that no keypair can reach, like zenoglyphs' burn sink. The account block's own fields do what the envelope did. `toAddress` identifies the protocol at zero data cost, the block's author is `msg.sender`, and NoM's account height replaces the nonce. A derivation node turns each momentum into one EVM block. It's Facet's design on NoM.

FOR

- An ERC-20 transfer is a type byte, a 20-byte target and 68 bytes of calldata: 89 bytes and 27,052 plasma, 25% less than option 3.
- A second tx type carries pre-signed EIP-2718 transactions, about 89 transfers per 16 KB block at ~12,750 plasma each. That's how MetaMask gets in, through relayers, as on Kasplex.
- Its consensus stands alone. EVM rules don't inherit zenoglyphs' rules, and zenoglyphs can change freely.
- It runs on a stock EVM client: a derivation node driving op-geth or op-reth over the Engine API. OP Stack's deposit transaction (type `0x7E`) already has the shape needed, with `from` set by derivation rather than by a signature. Solidity, Foundry, Blockscout and the bridge's own contracts all work unchanged.
- Contracts can read L1. The deriver sees every NoM block, so "pay on L1, settle on L2" is trustless and atomic, because both layers reorg together. It's the dispenser pattern, generalized.
- It survives its operators. The inputs live on NoM, so anyone can re-derive the chain.

AGAINST

- A new spec, a new node and new tooling. Option 3's viewer integration is lost, unless deploy-from-inscription is added.
- It shares option 1's gap: there's no trust-minimized way to take ZNN out without a counterparty, a committee or a spork.
- NoM's ordering becomes its ordering: address-sorted within a momentum, so priority can be ground with a vanity key generator (this tree has two).
- About 11 s to a soft confirmation, and 90 momentums (~15 min) to the finality the bridge assumes.
- Relayers must fuse QSR: 541 QSR for each 16 KB batch in flight.

CAN'T DO	Release L1 value without a counterparty, a committee or a spork. Confirm faster than NoM, unless a pre-confirming sequencer is added later.
VERDICT	Build this, on a stock client, with the batch lane from the start.

ADDED BY THIS PLAN

Routes the list was missing

None of these replace option 4. They're the pieces the prior art needed next to it: a way to go faster, a way to be verified, and three ways for value to leave.

ADDED · A

Batch lane now, sequencer later

Option 1 and option 4 converge. Keep the chain based, and later add an optional sequencer that pre-confirms and posts batches, with NoM as the path any user can force.

NEEDS	Nothing on NoM. A forced-inclusion rule in derivation.
WHEN	Once sub-second UX matters. Not v1.

ADDED · B

Validity proofs of the derived chain

Prove derivation and execution in a zkVM (the OP Succinct approach, or Reth under SP1 or RISC Zero) and publish the proofs.

NEEDS	Proving infrastructure to publish. A spork before NoM itself can verify.
BUYS	Light clients, bridges on chains that can verify proofs, and the future NoM verifier: Kaspa's Toccata path.

ADDED · C

The existing TSS bridge as the ZNN peg

Register the new chain as a third EVM network. The bridge administrator calls `SetNetwork`, each token pair passes a one-epoch time challenge, and the orchestrators add an RPC endpoint.

NEEDS The administrator and orchestrators to agree, and an EVM with `ecrecover`, which rules option 3 out.

TRUST Exactly what wZNN on Ethereum and BNB Chain relies on today.

ADDED · D

Counterparty exits: escrow on L2, pay on L1

A seller escrows an L2 asset. The buyer reserves it with a native tx, then pays the seller on L1 in a block that names the escrow, and the deriver releases it. It's dispenser rule R-7.4 generalized, and the reservation step closes the sold-out race dispensers have.

NEEDS Nothing on NoM. A system contract that shows contracts the L1 payments naming them.

TRUST None beyond NoM, but someone has to want the asset. HTLCs (the ferry and solzen tooling) cover other chains.

ADDED · E

Pillar attestations

Pillars or sentinels running the derivation node sign its state roots and publish them as NoM data. It's Igra's attesters, on Zenon's own validators.

NEEDS Nothing on NoM.

BUYS A confidence signal light wallets can check, and a ready-made signer set for route C.

ADDED · F

Enshrined execution by spork

Project Zeno's Settlement contract, an embedded ZK verifier, or an embedded EVM. This is the only route to trustless withdrawals.

NEEDS The governance hard fork in the draft ZIP (≥66% of pillars signalling, then a height at least 60 days out), and then a spork.

WHEN No date. Nothing in phases 0–4 should wait for it.

Not recommended: reviving Supernova. It means a second validator set with its own liveness, and the last one's operators have already left.

ELEVEN CRITERIA · FOUR OPTIONS

Side by side

CRITERION	1 · OPTIMISTIC ROLLUP	2 · ZENOGLYPHS	3 · EVM ON ZENOGLYPHS	4 · ZENOGLYPHS-LITE
Protocol change	✗ Its bridge and dispute game need a spork, which is blocked	✓ None	✓ None; one allowlist entry in zenoglyphs	✓ None
Who orders	✗ A sequencer	✓ NoM pillars	✓ NoM pillars	✓ NoM pillars; relayers choose what goes in a batch
Survives its operators	✗ No: it depends on the sequencer or a committee	✓ Yes: re-derivable from NoM	✓ Yes	✓ Yes
ZNN in	⚠ Through a bridge	— Not needed; dispensers are paid on L1	⚠ Burn deposit, designed but unbuilt (C5)	✓ Derived from an L1 send, as a burn receipt or through the bridge
ZNN out	⚠ A committee	— Not applicable	✗ None	⚠ Escrow swap (trustless) or the TSS bridge. A 1:1 peg needs a committee or a spork
EVM fidelity	✓ Stock client	— No VM	✗ Hand-written; shanghai subset, no precompiles	✓ Stock client
Wallets	⚠ MetaMask only	✓ Syrius	⚠ Syrius only	✓ Syrius, and MetaMask through a relayer
Plasma per ERC-20 transfer	✓ Lowest: compressed batches	— —	✗ 36,232	✓ 27,052 native · ~12,750 batched
Ceiling with all of NoM	✓ Limited by execution, not NoM	⚠ ~9 ops/s	✗ ~9 calls/s	✓ ~9/s native · ~790/s batched
Consensus coupling	⚠ The stack assumes Ethereum throughout	✓ Self-contained	✗ Inherits every zenoglyphs rule	✓ Self-contained

CRITERION	1 · OPTIMISTIC ROLLUP	2 · ZENOGLYPHS	3 · EVM ON ZENOGLYPHS	4 · ZENOGLYPHS-LITE
Built today	✗ Nothing	✓ v1 frozen, running on devnet	⚠ Devnet prototype	⚠ Nothing yet; reuses option 3's lessons and code

MAINNET RULES · COMPUTED FROM GO-ZENON CONSTANTS

What one ERC-20 transfer costs

Plasma is capital, not a fee: fused QSR comes back every momentum. So the figures below are what you have to lock up, or grind, to send at once. Batching is what changes the picture, not the envelope.

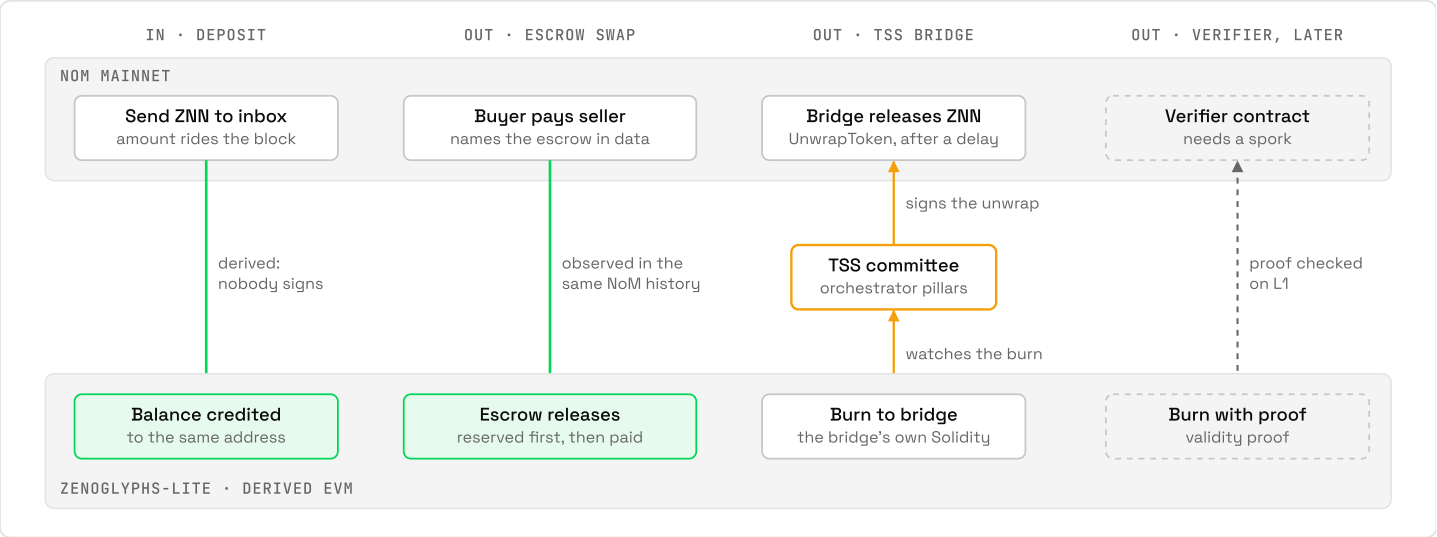
DESIGN	BYTES ON L1	PLASMA	QSR FUSED	OR PROOF OF WORK	TRANSFERS PER BLOCK
Option 3 · inscription 156 B envelope + 68 B calldata	224	36,232	18	54.3 M hashes · ~54 s	1
Option 4 · native tx type + 20 B target + 68 B calldata	89	27,052	13	40.6 M hashes · ~41 s	1
Option 4 · batched signed EIP-1559 tx, ~183 B each, 16 KB block	~183 each	~12,750 each	541 / block	1,080 B cap: 5 txs · ~2 min	~89

- **Ceiling:** if one design took all of NoM (100 blocks per ~11.25 s momentum), one tx per block tops out at 8.9 tx/s. Batched blocks reach about 790 tx/s, and execution limits it well before that.
- **The cost that lands on others:** every sustained 1 KB/s of rollup data adds about 86 MB a day to every NoM node, before indexes. Using the full 100 × 16 KB per momentum would add about 12.6 GB a day to each. So derivation should cap the rollup's footprint (decision 5). Otherwise NoM node operators pay for its success in disk.
- **Proof of work** figures assume the ~1 MH/s solzen measured in Go. Fusing about 13 QSR removes the wait for a native tx entirely.

OPTION 4 · THE RECOMMENDED SHAPE

How value moves in and out

A derived chain can see everything on NoM, and NoM can see nothing of it. That asymmetry decides which directions are free.



Value enters for free, because every deriver sees the L1 send (green). It leaves only three ways. A counterparty can pay on L1 in the same history the escrow releases in. A committee can sign for the bridge (amber), which is the trust wZNN already relies on. Or a verifier contract can check a proof, and that needs a spork NoM can't currently activate (dashed).

What the derivation node does

This is enough to write the spec in phase 0. Each line is a rule every node has to apply identically.

INPUTS	Blocks sent to the inbox address. Bridge `WrapToken` calls and L1 payments that name an escrow are read as L1 facts. Everything else is ignored.
TX TYPES	`0x01` native: `rlp([to, value, gas, data])`, sent by the block's author, with no signature; the NoM height is the nonce. `0x02` batch: signed EIP-2718 txs, posted by a relayer. `0x03` chunk: raw init-code bytes. A deploy lists chunk block hashes, which is zenoglyphs' refs pattern: independent blocks published 13 ms apart, where chained parts cost ~30 s per link.
BLOCKS	One EVM block per momentum, with the momentum's timestamp. An empty momentum becomes an empty block (Facet's filler rule).
GAS	A native tx's budget scales with the plasma its block paid. The prototype uses 8 gas per plasma; BRC2.0 uses 12,000 gas per byte. A per-block ceiling skips deterministically (`evm-todo` R1). Relayed txs pay gas in the chain's own currency.
FINALITY	Soft on inclusion in a momentum, final at 90 momentums, the number the bridge already uses. A NoM reorg rewinds the EVM through `engine_forkchoiceUpdated`.
OPEN RULES	Order inside a momentum, and the footprint cap. Both are yours to decide (decisions 4 and 5).

PATH A · IN ORDER · EACH PHASE ENDS ON A TEST

The plan

Sizes are relative, for someone who already knows this tree: S is days, M is weeks. Phases 0 to 4 need nothing from NoM governance. Phase 5 does, and has no date.

0 Decide, then freeze the envelope SIZE S

BUILD

- Answer the round-trip question (Path A or B) and decisions 2–6.
- Write `spec/LITE.md`: the inbox address, tx types, deposit rule, gas and per-block ceiling, order rule, footprint cap, chain id and pinned EVM fork.
- Golden vectors from day one. The prototype never had EVM vectors (`evm-todo` 04).

DONE WHEN

The spec and vectors are reviewed, and every decision is recorded with what reversing it costs, the way `ASSUMPTIONS.md` does it.

⚠ If the answer is Path B, phase 1 becomes BRC2.0's shape instead: revm or geth's `core/vm` inside zcribe in place of the interpreter, plus a `transact` op and zenoglyph deposit and withdraw rules.

1 A derivation node on devnet SIZE M

BUILD

- A Go derivation node that reads momentums with `ledger.getDetailedMomentumsByHeight` (1,024 per call), reuses zenoglyphs' RPC client and reorg handling, and drives op-geth or op-reth over the Engine API with `noTxPool`.
- Native txs become deposit-style transactions: `from` is the NoM author, and the source hash is the NoM block hash.

DONE WHEN

Two independent nodes agree on every block hash and state root through a forced reorg. The prototype's swarm replays as lite txs. A stock Uniswap v2 pair and an ERC-721 deploy unmodified.

2 Wallets SIZE M

BUILD

- The Syrius path: a browser tx builder reusing `web/ui/src/wallet/` (bech32 with a checked checksum, keccak), with RLP in place of CBOR.
- The MetaMask path: a relayer that batches signed txs into NoM blocks from fused QSR and charges in the chain's own currency.
- Blockscout on the node's JSON-RPC, in place of a custom explorer.

DONE WHEN

A MetaMask user with no Zenon key and a Syrius user with no EVM key both complete the same swap.

3 Value in and out — SIZE M

BUILD

- Wire deposits for the currency chosen in decision 2.
- The system contract that shows contracts the L1 payments naming them, plus the escrow exit (route D) with its reservation step.
- Optionally, one-way deposits of zenoglyph units.

DONE WHEN

An L2 asset sells for L1 ZNN with no custodian, including the refund after a missed payment window.

4 Mainnet — SIZE S, PLUS A SOAK

BUILD

- Pick the start height, and publish the spec, vectors and binaries.
- At least two independent operators, and a compare tool that bisects to the first divergent block, as `zcribe compare` does.
- Optionally, pillar attestations (route E).

DONE WHEN

Independent nodes agree for 30 days, and the soft and final labels are documented for wallets.

5 What governance unlocks — NO DATE

BUILD

- Ask the bridge administrator to register the chain (route C). That's an admin action, not a spork, but it needs the orchestrators to agree.
- Validity proofs of derivation and execution (route B).
- Once a governance contract exists: a spork for a verifier or settlement contract (route F). This is the step that makes exits trustless.

DONE WHEN

A withdrawal settles on NoM with nothing but a proof behind it.

NEEDED BEFORE PHASE 0 CLOSES

Decisions only you can make

1. **Round trip or one way.** Must zenoglyph assets come back out of contracts? Yes means Path B, a merged indexer like BRC2.0. No means Path A, which is what this plan assumes.
2. **What ZNN is inside.** A burn receipt: trustless, but never redeemable 1:1. A TSS-bridged token: redeemable, with orchestrator trust. Or no ZNN at all, and a closed native currency. Gas works without any of them, because native txs buy their gas with plasma.
3. **MetaMask in v1, or Syrius first.** MetaMask needs relayers holding fused QSR and a currency to pay them in. It's also how every comparable chain found users.
4. **Order inside a momentum.** Keep NoM's address sort, which anyone with a vanity key generator can game, or shuffle by the momentum hash, which only the pillar producing that momentum can game.
5. **Footprint cap.** How much of NoM's 100 blocks per momentum, and of every node's disk, the chain may use. It's enforced in derivation, so blocks over the cap are wasted and relayers stop sending them.
6. **Execution client.** op-geth, Facet's choice, in Go like the rest of this tree. Or op-reth, Igra's, in Rust.

OPEN, WITH WHAT WOULD SETTLE EACH

Risks and open questions

- **The governance freeze.** The draft ZIP adds a governance contract by hard fork: at least 66% of pillars signal, then a height at least 60 days out is set. It has no date, and route F waits on it.
- **Reorg depth.** 90 momentums is the bridge's working assumption, not a measured bound. Inside NoM and the chain it doesn't matter, because both reorg together. It matters for anything that leaves: bridges and exchanges. Measure mainnet fork depth before sizing those delays.
- **Pillar-level MEV and censorship.** When more than 100 blocks are waiting, the producing pillar decides which get in. Before dynamic plasma that selection walks a Go map, so it's effectively arbitrary. A based chain inherits this.
- **Dynamic plasma arriving later.** If the spork ever activates, the price and the wedging measured on devnet arrive too. Relayers will need price headroom and rebids. `internal/inscribe` already has both.

- **Operators aren't paid.** Nothing in the protocol pays the people who derive the chain (`evm-todo` C3). Charge for RPC and attestation, not gas.
- **Demand.** Mainnet carries 0.09 blocks per momentum, and Facet's Stage 2 chain holds about $650K. Build for the assets and wallets people already have.

READ 11 SEP 2026

Sources

- BRC2.0 programmable module (Best in Slot)
- BRC2.0 goes live at block 912,690 (CoinDesk)
- Facet documentation
- Facet on L2BEAT
- Facet VM launch, Nov 2023 (CoinCarp)
- Kasplex L2, a based rollup on Kaspa (KEF)
- Kasplex zkEVM mainnet (Barchart)
- Kasplex L2 docs
- Igra public mainnet (Chainwire)
- Kaspa Toccata activation record
- Alkanes (GitHub)
- Counterparty EVM mainnet proposal and forum thread
- Supernova extension chain, soft launch, repository
- Project Zeno (zenon-developer-commons)
- Draft Governance ZIP (HyperCore One forum)
- IOTA EVM (IOTA docs)
- This tree: `projects/zenoglyphs/research/` (evm-prototype, evm-benchmark, evm-fee-design, evm-address-scheme, evm-todo, on-chain-execution), `spec/SPEC.md` , `ASSUMPTIONS.md` A-42, and the local `zenoglyphs.db` for the 8,464-call envelope measurement
- go-zenon: `vm/constants/plasma.go` , `vm/plasma.go` , `chain/account_pool.go` , `chain/nom/momentum_content.go` , `vm/embedded/definition/spork.go` , `vm/embedded/implementation/bridge.go` , and `docs/DYNAMIC_PLASMA.md`
- Mainnet RPC `node.zenonhub.io:35997` : `ledger.getFrontierMomentum` , `ledger.getMomentumsByHeight` , `embedded.spork.getAll` , `embedded.bridge.getAllNetworks` , `getOrchestratorInfo`

PREPARED 11 SEP 2026 · MAINNET FIGURES CAN BE RE-READ WITH THE RPC CALLS ABOVE · DEVNET FIGURES SAY SO WHERE THEY APPEAR